ScienceDirect

Journal of Power Sources

Volume 626, 15 January 2025, 235729

Thermal strain offset strategy for fabrication of glass-ceramic to metal seals with high-reliability

Ming Huang, Fenglin Wang, Haijun Mao, Zhuofeng Liu, Wei Li, Weijun Zhang, Xingyu Chen

Show more

Outline | Share Cite

https://doi.org/10.1016/j.jpowsour.2024.235729

Get rights and content

Full text access

Highlights

- Flexible Thermal Strain Offset Strategy.
- Efficient Thermal Treatment Process for high-expansion metal.
- Optimized Glass-Ceramic Material with High Thermal Stability.

Abstract

Excessive thermal stress in the glass-ceramics to metal seals (GCtM) is known to be inimical to the reliability of the related electrical components and thus typically avoided during the sealing process. Quartz, a linear high-expansion material, has long attracted extensive attentions in GCtM seals, ascribed to its ability to ameliorate the thermal stress in the sealing component. However, the quartz crystallization in the lithium silicate recrystallization glass-ceramics need for a series of complex and cumbersome thermal profiles, especially including a rapid cooling process (>70°C/min), in the conventional method. Here, we disclose a key finding that a small amount of quartz seed could promotes quartz crystallization at a slower cooling rate and higher temperatures. To counter the problem mentioned above, a thermal strain offset strategy is proposed, which embodies a model to predict the thermal expansion behaviors at different temperatures and the controlled growth of quartz. Consequently, a novel glass-ceramics with near-linear thermal strain have been obtained, notably through a facile and high-efficient thermal profile. More surprisingly, the improved sealing glass exhibits a fluidity similar to that of the original material and the distinctive ability to ameliorate the thermal stress at elevated temperature. This work sheds light on developing high-reliable and cost-effective manufacture method for industrial production of the related electronic component.

Previous

Next

Keywords

Glass-to-metal seals; Quartz growth; High-efficiency thermal schedule

1. Introduction

As electronics industry has developed by leaps and bounds over the past decades, the systems have become more and more complex, such as manned rocket, remotely piloted vehicle, and nuclear power stations. Higher requirements are put forward to the higher reliability electronic component to avoid the occurrence of fatal problems in engineering projects [[1], [2], [3]]. Nevertheless, some of gas types, like oxygen and steam, may pose a serious threat to the large-scale electronic system and thus the hermeticity is one of the most importance performances in the manufacture process of some electronic components to divide the sensitive part in the system from the external environment [[4], [5], [6], [7]]. More specifically, hermeticity refers to the ability a sealed system to prevent the entry or exit of gases, liquids, or particles. Meanwhile, matched seals, one of the sealing technologies, are often used for highly reliable hermetic electronic packages because of the advantages of being stress-free and temperature-resistance, but this method requires that the difference between coefficient of the expansion thermal (CTE) values of sealant and metal materials must be within 10% to prevent the thermal stress accumulation [[8], [9], [10]].

Glass-ceramics have been considered promising for their potential in the hermetic sealing technology because of low melting point, tunable CTE value and excellent mechanical performances. Therefore, the glass-ceramics is one of the most promising candidates to realize the matched seals with high-expansion alloy. The thermal expansion of the glass-ceramics is largely depended on what and how many crystalline phases the glass-ceramic contains. High-expansion characteristic in glass-ceramic system generally arise from the precipitation of cristobalite, quartz, $Ba_{1-x}Zn_xSi_2O_7$, $Nd_{1-x}Si_xMnO_{3\pm\delta}$, $SrCo_{0.8}Fe_{0.2}O_{3-\delta}$ [[11], [12], [13]] et al. In order to obtain proper type and content of crystalline phases in glass, a thermal profile need to be carefully designed. For example, for the formation of cristobalite in the lithium silicate system, it is essential to maintain a temperature above 950°C for a while during the sealing process, otherwise the crystallization of this polymorph will not occur upon the subsequent cooling. Among all types of the glass-ceramics used for the hermetic sealing, the recrystallization lithium silicate glass-ceramic attracts significant attention due to their special physical-chemical properties, such as high thermal expansion (~16.0 ppm/°C), excellent electrical resistance and good mechanical strength. These properties, particularly the thermal expansion, promote the application of the lithium silicate glass-ceramic on a matched hermetic sealing with high-expansion alloys. In this glass-ceramic system, the lithium metasilicate (CTE: 13.0 ppm/°C) and cristobalite (CTE: 27.1 ppm/°C) phases crystallized on the subsequent sealing process result in the high thermal expansion of this glass-ceramic [[14], [15], [16], [17]]. However, α-β phase transition occurs in the cristobalite at the temperature range of 200~300°C, accompanying with a large volume change, which leads to a nonlinear change in the thermal strain of the aforementioned glass-ceramics. The step-like thermal strain apparently will lead to a thermal expansion mismatch between the sealant and metal materials on account that the temperature dependency of the thermal expansion of metal materials usually is linear. It is generally known that the stress induced by the mismatch of the thermal strain curve is fatal to the reliability of the electrical components. Cracks and gas leakage may occur in these components, which could result in the collapse of the entire system, owing to the thermal mismatch [18,19].

To tackle this issue, it is highly desirable to develop viable approaches to moderate the thermal strain of the glass-ceramics to match the thermal expansion characteristics of metal materials. One possible route to eliminate the mismatch is to introduce a material or phase with an opposite thermal strain curve into the glass-ceramics to offset the step-like change of the thermal expansion. For example, Steven et al. [16] proposed a novel thermal profile to promote the quartz crystallization, and one of the key steps in the profile included the rapid cooling (>70 K/min) after the sealing peak temperature to avoid the crystallization of cristobalite. Although this method successfully reduced the degree of the nonlinearity of the thermal strain, there were still several problems involved in the industrial production and the large-scale electronic system, such as higher requirements for manufacturing facility, lower production efficiency and lower sealing reliability. However, the addition of the quartz seed from the external increased the crystallization temperature of quartz up to the temperature that cristobalite forms, at about 800~900 °C, in the previous research [16].In other words, the sealing thermal profile of the lithium silicate glass-ceramics is extremely simplified, and the sealant as well as the metal do not have to experience the rapid cooling, indicating that the interface between the two kinds of materials will accumulate less thermal stress and exhibit the reliable hermeticity performance. Thus, it is undoubtful that fundamental understanding about the interaction between the quartz seed and glass should help us to uncover the route of preciously manipulating the thermal expansion behaviors for advancing the facile fabrication toward highly reliable electronic components. Obviously, this method of directly sealing with high-expansion metals can also provide a new technological route for medium and low-temperature solid oxide fuel cells [7,20]. For a long time, sealing of solid oxide fuel cells has been challenging due to the difficulty in finding suitable sealing materials, often resorting to brazing to seal the cells. However, metal solder is highly susceptible to failure in the long-term high-temperature oxidation environment, leading to a decrease in the efficiency of sold oxide fuel cells and even damage [21,22].

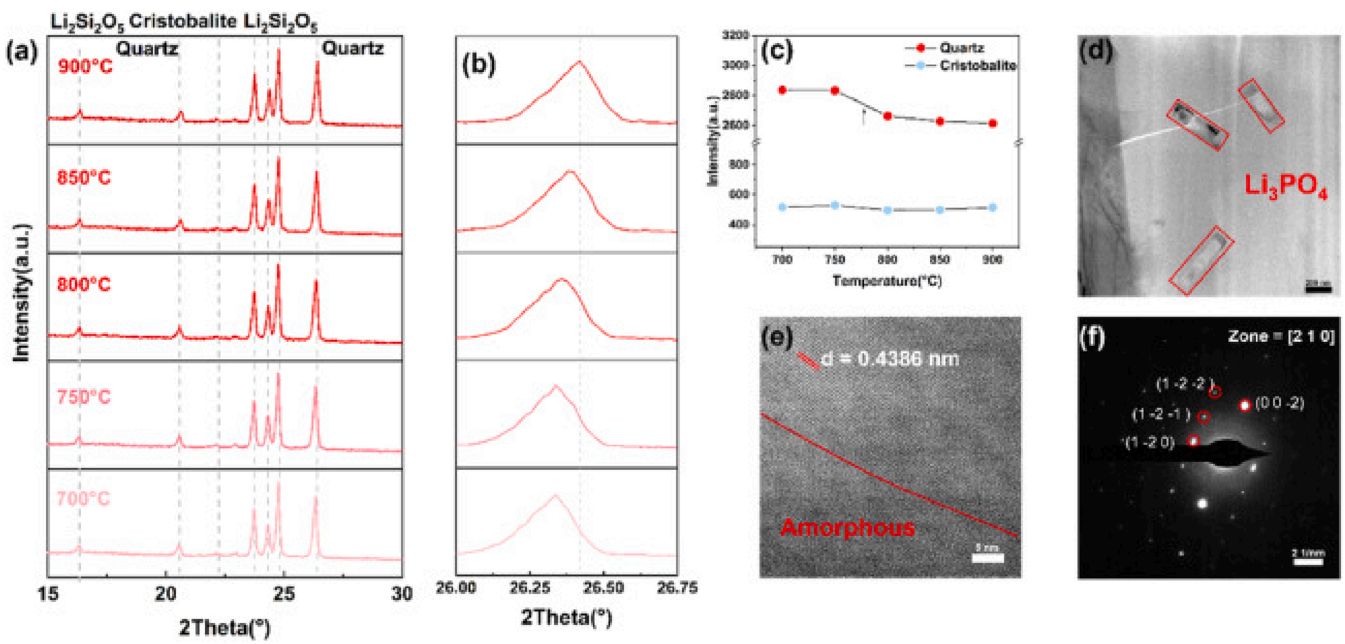
Herein, we present a thermal expansion offset strategy for alleviating the mismatch induced by the α-β phase transformation of cristobalite, which in effect improve the reliability of the electrical feedthroughs. The strategy was developed based on the fundamental understanding of the predicting model of the thermal expansion coefficient of particle-reinforced composites. Here, the glass-ceramic materials were considered as particle-reinforced composites. Only two types of crystalline, cristobalite and quartz, were regarded as the particles in the composites, and would be taken into the consideration of the model construction. Based on the Ockham's Razor theory, the difference among lithium metasilicate, lithium phosphate and amorphous phase was ignored, and three crystalline phases were regarded as the matrix phase in the composites. The calculation results of this model allowed us to know the influence mechanism of quartz content on the thermal strain of the composites, thus guided us to obtain the optimized concentration of quartz. The employed scenario, wherein a small amount of quartz can be used to trigger the quartz crystallization in the glass-ceramics at a specific temperature range, is significantly different from what has been reported previously [16]. Thus, there is no need for designing cumbersome and difficult thermal profile to induce the occurrence of quartz.

To improve the reliability of the metal-to-glass sealing, we herein propose a thermal expansion offset strategy, which embodies the analytic model for the influence of quartz-cristobalite ratio and the introduction of the quartz seed on the thermal strain and the coefficient of thermal expansion, making for the crystallization of the quartz and cristobalite in the glass-ceramics at the same temperature. In this way, the thermal strain curve of this glass-ceramic has been well tuned to alleviate the thermal expansion mismatch due to the β-to α-cristobalite phase transformation. More importantly, this innovation has led to a conspicuous performance improvement for the reliability of the metal-glass sealing and reduce the complexity as well as energy-time consumption of the thermal profile compared with the traditional heat-treatment procedures, which is crucial advantages required for the industry.

2. Result and discussion

2.1. Quartz seed promotes quartz crystallization at elevated temperature

The induction of the quartz crystallization is not an easy step according to the previous reports [16].Especially, the induction process should be included into the sealing profile and should not influence the quality of the sealing component at the same time. In the traditional method, the crystallization of quartz phase only occurs when the sample experiences a rapid cooling, faster than 50°C/min, from the sealing peak temperature to 650°C. However, the actual requirements for the cooling rate, usually larger than 70°C/min, are much stricter than the reported one in our earlier experiments [15]. Apparently, if the crystallization of quartz occurs in the above way, there are several problems, such as the precisely control of the cooling rate and the accumulation of the thermal stress in the rapid cooling. Herein, a novel quartz seed is introduced into the lithium silicate recrystallization glass-ceramics (LS-GC) to lower the requirements of the quartz crystallization. The LS-GC with the addition of 7 wt% quartz seed firstly was heated up to 1000°C and held for 10 min to simulate the real thermal treatment schedule. After the end of sealing temperature, the sample went through a cooling process at a rate of 20°C/min. The sample was scanned by X-ray, during 900~700°C at the interval of 50°C during the cooling process. Fig. 1a are the diffraction patterns of the sample respectively at 900°C, 850°C, 800°C, 750°C and 700°C. It's obvious that the phase composition of sample at different temperatures don't change obviously, and mainly consists of lithium disilicate, quartz and a few cristobalite. Theoretically, the phase composition in the glass-ceramics should only contain lithium silicate, quartz and cristobalite, thus the occurrence of lithium disilicate is possibly ascribed to the change of the thermal profile because of the time consumption in the X-ray scanning. Fig. 1b is the (011) peaks of the quartz phase, of which the position clearly shifts toward the lower degree, suggesting that the quartz phase exists in a tensile stress environment and thus the interplanar spacing of the quartz have been increased. The glassy phase has a lower thermal expansion than that of the quartz phase and that is to say, the volume change of the quartz crystalline phase is larger than that of the glassy phase. Consequently, the quartz would be pulled by the glassy region once the environment temperature decreased.



Download: Download high-res image (607KB)

Download: Download full-size image

Fig. 1. **Quartz seed accelerates the formation of quartz at relatively slower cooling rate**. **a** The High-Temperature XRD (HTXRD) patterns of the glass-ceramic with the addition of 7 wt% quartz seed at 900 °C, 850 °C, 800 °C, 750 °C and 700 °C when the sample was in a cooling process with a rate of 20 °C/min **b** The magnified pattern of (011) peak corresponding to quartz phase in the glass-ceramics. **c** Absolute intensity of (011) and (101) peaks respectively corresponding to the quartz (red dot) and cristobalite (blue dot). **d** The transmission electron microscope (TEM) and **e** high-resolution transmission electron microscope (HRTEM) images of the glass-ceramics with the addition of 7 wt% quartz seed. **f** The selected electron diffraction pattern of the columnar crystal (Li_3PO_4) in the TEM image. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Although the thermal treatment schedule has been changed due to the crystallization phenomenon during X-ray scanning, the intensity of quartz and cristobalite phases still reflect the influence of quartz seed on the crystallization process of the high-expansion phase. As shown in Fig. 1b, the intensity of (011) quartz has increased at the middle of the 700–800 °C. More importantly, the introduction of quartz seed seemed to suppress the formation of cristobalite, for the intensity of (101) peak corresponding to cristobalite nearly maintains a constant. TEM image of sample is shown in Fig. 1d and the areas highlighted by red rectangles should be corresponded to lithium phosphate (Li_3PO_4), with a columnar shape, which could be demonstrated by the distribution of phosphorus element, as shown in supplementary. It is clear that in the TEM image, the ends of the lithium phosphate don't have the evidence of the cristobalite growth, while there should have been occupied by cristobalite crystals without the influences of the quartz seed, according to the previous reports [23]. Additionally, the area of lithium phosphate could be divided into two regions, the crystalized and amorphous regions, as shown in Fig. 1e and f. Of particular note that crystalized Li_3PO_4 (PDF number: 04-007-2815), with a certain shape, is extensively thought as the nuclei for the growth of cristobalite phase [19,20]. Therefore, this suggested that the essence that the quartz seed hinder the cristobalite crystallization may slower the crystallization rate of lithium phosphate. All in all, quartz seed might be crucial for the fabrication of near-linear strain sealing glass-ceramics, thus become a key role for high-reliability sealing component.

2.2. How quartz-cristobalite ratio influences the thermal strain behaviors of the glass-ceramics

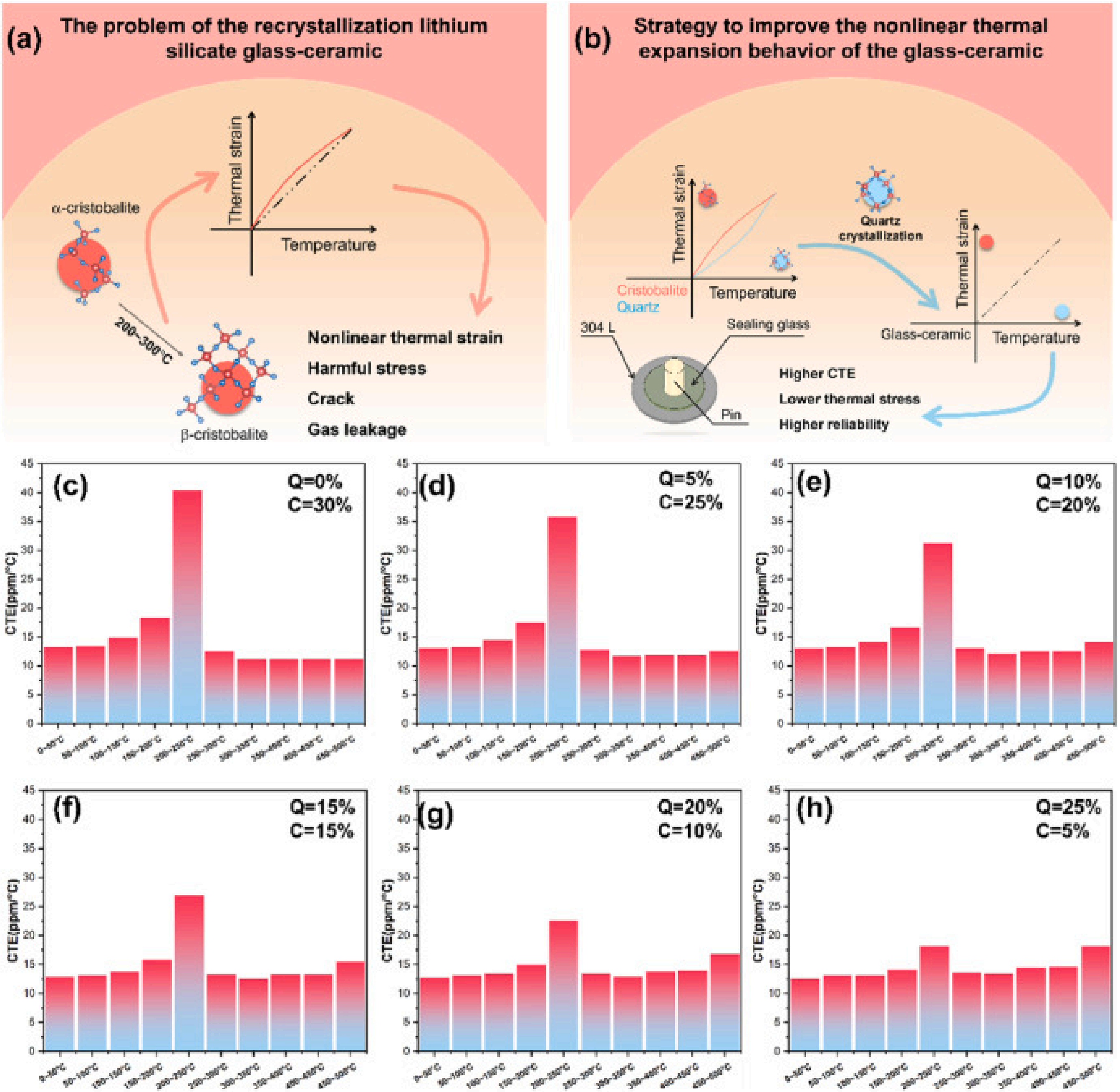
Conventional recrystallization lithium silicate glass-ceramics, of which the thermal expansion reaches up to ~16.0 ppm/°C, need to experience a special thermal profile, containing several temperature ranges for nucleation and cristobalite growth. Nevertheless, coins has two sides. On the one hand, cristobalite have an ultra-high coefficient of thermal expansion (CTE) and thus become the main source of high-expansion of the glass-ceramic. On the other hand, α-β phase transition between low- and high-cristobalite results in a nonlinear change in the thermal strain curve of this glass-ceramics. Obviously, the step-like curves may lead to harmful inner stress and even cracks at the interface, eventually causing the gas leakage of the component (Fig. 1, left). For the problem mentioned above, quartz phase was considered as a thermal expansion offset phase to alleviate the thermal stress due to the mismatch (Fig. 1,right). To understand the relationship between quartz crystallization and the thermal strain behaviors of the glass-ceramics, a model based on thermoelastic mechanics, of which the original function is to calculate the CTE of particle-reinforced composites, have been applied to interpret the question, as shown in the following equation:

$$\alpha_c = \sum_{i=1}^{n} \frac{f_i K_i \alpha_i}{4G_c+3K_i} / \sum_{i=1}^{n} \frac{f_i K_i}{4G_c+3K_i}$$

$$E_{\mathrm{c}} = \sqrt{\sum_{i=1}^{n} f_i E_i / \sum_{i=1}^{n} \frac{f_i}{E_i}}$$

Where α_c, E_{c}, G_{c} represent the coefficient of the thermal expansion, elastic modulus, and shear modulus of the composites, respectively. E_i, K_i, f_i denote the elastic modulus, bulk modulus, and the volume fraction of the i_{th} phases in the composites, respectively. J.W.P. Schmelzer [24] reported that modulus of a glass is a function of temperature and only slightly decreases under the glass transition temperature. In this model, therefore, the modulus is considered to be a constant and will not vary with temperature [24]. The paraments included in the model could be found in the supplementary. Meanwhile, the percentage of amorphous and crystalline phase in the recrystallization lithium silicate glass-ceramic in the literature [16] were used to verify of the accuracy of this model and the results shown that the coefficient of thermal expansion derived by this model at 0–500°C is 15.91 ppm/°C, closer to the experiment values (16.11 ppm/°C). In short, the consequence of such model shows excellent ability to describe the thermal expansion behavior of aforementioned glass-ceramics.

To make the point about how quartz content influences the thermal strain of the glass, the whole temperature range were homogenously separated into ten parts at the interval of 50°C, and then the average thermal expansions were formulated at some specific parts. Fig. 2(b)–(g) shows the thermal expansion behavior of the composites with different quartz concentrations at 0–500°C. Overall, there has been a surge of thermal expansion at the fifth parts (200–250°C) in all samples. And it's the main reason why a high-expansion sealant containing cristobalite have lower reliability. Meanwhile, it was demonstrated that the increase in quartz concentration were beneficial to optimize the nonlinear thermal expansion behavior of the glass-ceramics. In other words, with the increase of the quartz-cristobalite ratio, the thermal expansion behavior of the composites in the ten parts is gradually closer to a constant, suggesting that the increment of the quartz-cristobalite ratio in the materials will be effective to relieves the mismatch stress. It's important to point out that, while the quartz phase exhibits a marvelous performance to reduce the nonlinear degree of the thermal strain curve of the glass-ceramics, the increase of the quartz simultaneously decrease the coefficient of thermal expansion. Accordingly, to fabricate a glass-ceramics with high thermal expansion and near-linear thermal strain, the design of phase composition concludes that the quartz-cristobalite ratio in glass-ceramics should be closer to 1:1 and at the same time, the cristobalite concentration should be guaranteed at the highest possible level.

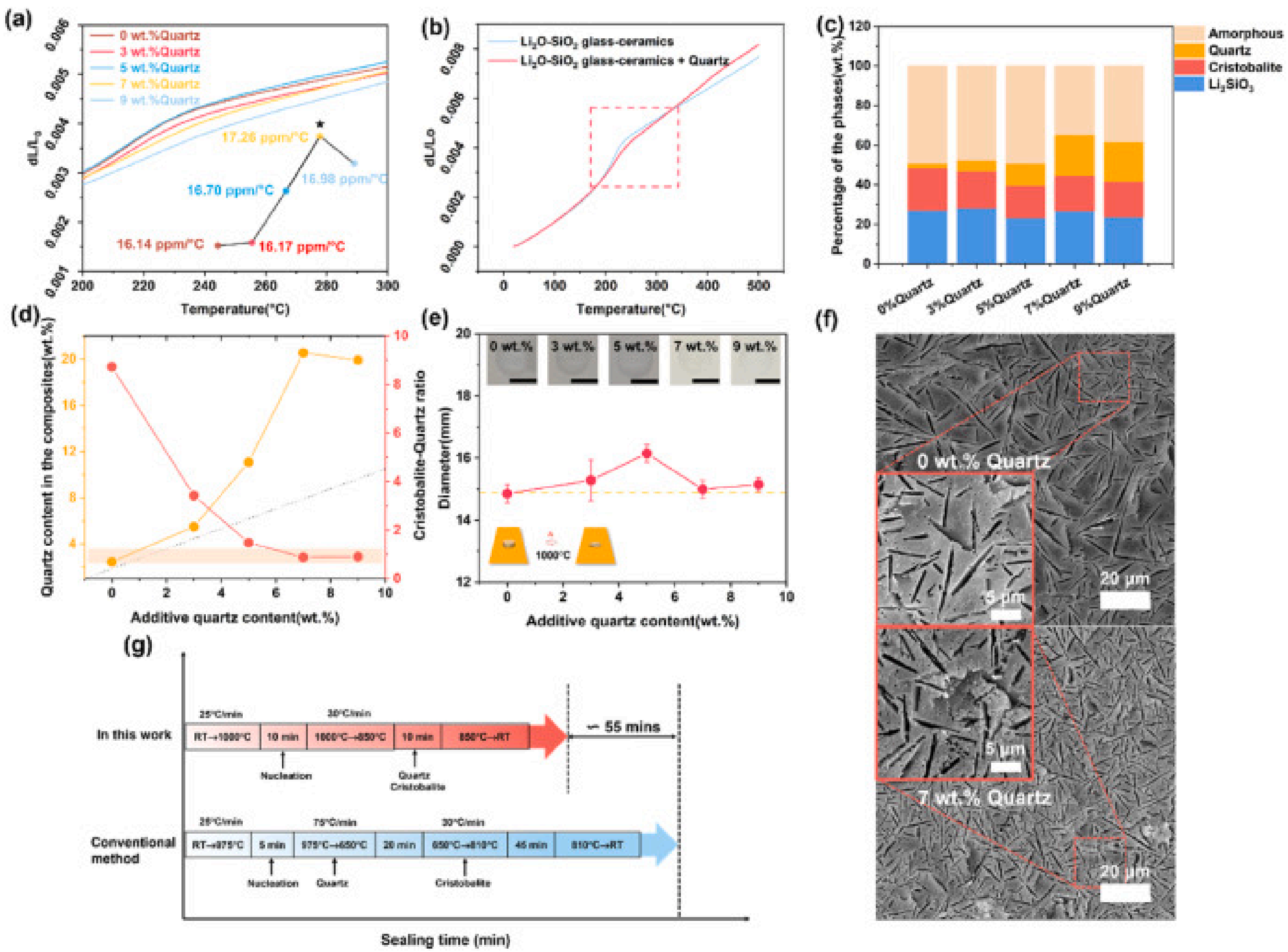


Download: Download high-res image (1MB)

Download: Download full-size image

Fig. 2. **Quartz alleviates the nonlinear thermal expansion behavior of the recrystallization lithium silicate glass-ceramic. a** The problem of crystallization lithium silicate glass-ceramic and the schematic diagram of the strategy to reduce the nonlinear thermal expansion behavior of the glass-ceramics. **b** Strategy to improve non-linear thermal expansion behavior of the glass-ceramic. The calculated coefficient of thermal expansion of the glass-ceramics based on the model with the quartz concentration of **c** 0%, **d** 5%, **e** 10%, **f** 15%, **g** 20%, **h** 25%. The higher quartz content represent that the thermal expansion strain curve is closer to linearity while the values of the CTE are smaller with the increasing concentration of quartz. **Q** and **C** denotes the quartz and cristobalite concentration in the glass-ceramic, respectively.

Until now, it's clear how the quartz crystallization was induced and how many quartz should the glass-ceramics contains. Thus, different concentrations (0–9 wt%) of the quartz seed were introduced into the glass-ceramics and Fig. 3a is the thermal strain curves of the samples at 200–300 °C. The linear degree of thermal strain curve becomes higher with the increase of quartz seed concentration. The CTE of the samples reach up a maximum value, 17.26 ppm/°C when the addition content of quartz seed is 7 wt%. But once the concentration of the quartz seed exceeds 7 wt%, the thermal expansion coefficient will decrease, which is basically consistent with the result predicted by our model. The comparison of the thermal strain curves with and without the quartz seed, as presented in Fig. 3b, demonstrates that our offset strategy is effective to suppress the non-linear thermal strain curve of the glass-ceramics. Meanwhile, due to the massively quartz crystallization, the thermal strain curve of glass-ceramics at the elevated temperature maintains a relatively large slope compared to the original glass-ceramics, roughly speaking, which represents a better transition between the glass-ceramics and metal materials. To quantitively evaluate the phase composition of samples, an internal standard method, employing α-Al_2O_3 as standard material, is applied to determine the percentage of amorphous and crystalline phases in the glass-ceramic with the addition of quartz seed. The improved and original sealing glass have the same phase compositions, including amorphous glass, Li_2SiO_3, cristobalite and quartz, as shown in Fig. 3c. The main differences among the samples are the quartz content. It's obvious that the quartz content in the improved glass-ceramic have been increased as the quartz seed concentration grown. More importantly, Fig. 3d demonstrates that the slight addition of quartz seed triggers the crystallization of the more quartz in the glass-ceramics. In general, the excessive quartz from the external environment severely leads to the deterioration of the fluidity of the sealing glass, which is one of the most important properties for the sealing process. Consequently, at the elevated temperature, the glass-ceramics should fill the gap between sealing glass and the metal materials to ensure the hermetic of the sealing component. To further measure the influence of quartz seed on the fluidity of glass-ceramics, button tests are carried out. The average diameter of the samples after the heat-treatment will be an indicator for the evaluation of the fluidity of glass-ceramics with the different additions of the quartz seed. The results, as shown in Fig. 3e, suggest that the fluidity of all samples maintains a good level at the condition that concentration of the quartz seed is under 10 wt%. Specifically, the curve of the average diameter of the samples looks like a “mountain peak” and it reaches up to the maximum values when the concentration is 5 wt%. And after the peak, the fluidity of the glass-ceramics doped with the quartz seed starts to mildly deteriorate. Fig. 3f are the SEM images of the glass-ceramics doped with 0 and 7 wt% quartz seed after HF corrosion. There are a lot of the needle-like holes on the samples surface, and the formation of these holes are mainly due to the active reaction of the lithium silicate with HF solution. And the areas highlighted by red rectangles demonstrated that the quartz seed also slightly influences the densification of glass-ceramics, since the corroded surfaces of the sample doped with the quartz seed exhibits more connection points than that of the original sealing glass. To sum up, it's concluded that the quartz seed could effectively ameliorate the mismatch due to the non-linear thermal strain curve of the lithium silicate recrystallization glass-ceramics. At the same time, as shown in Fig. 3g, the sealing time have been largely reduced, about 55 min, indicating our offset strategy simultaneously enhancing the efficiency in the production of sealant. Most of the reduced time should be attributed to the stages that are designed for the induction of the cristobalite and quartz crystallization, compared to the conventional method.



Download: Download high-res image (976KB)

Download: Download full-size image

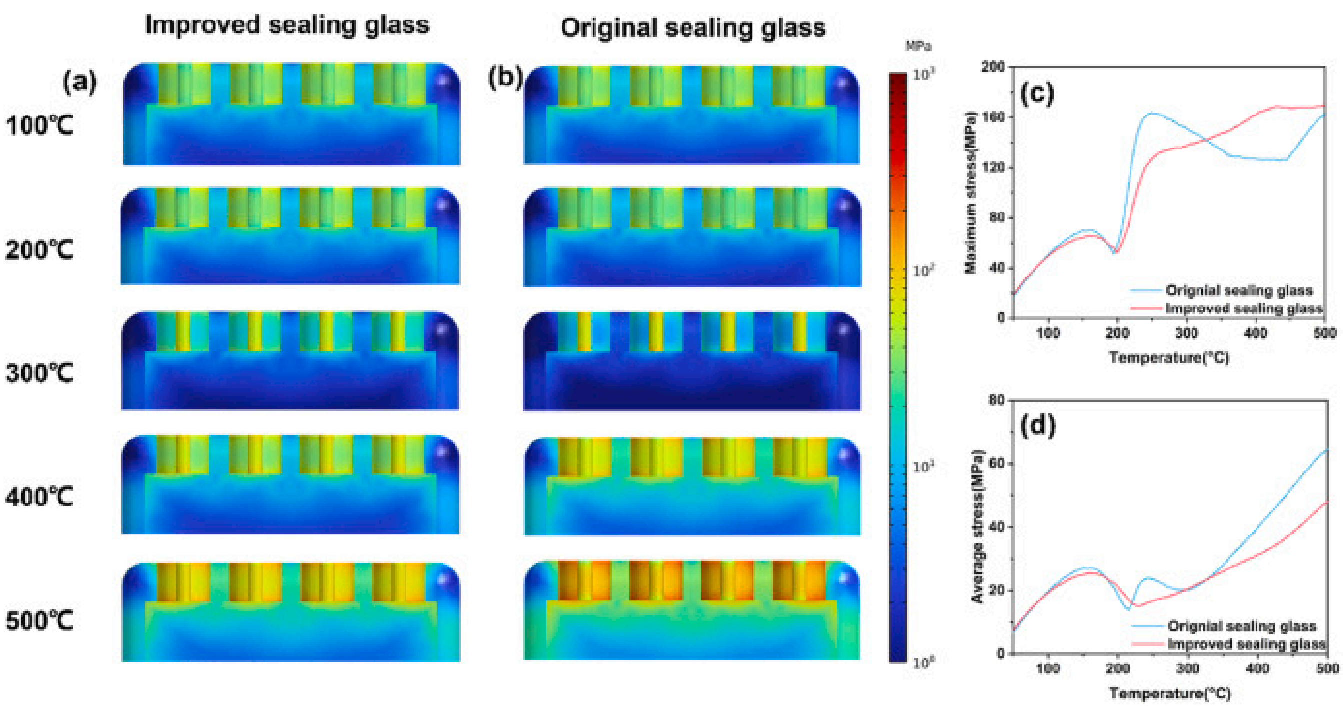
Fig. 3. **The thermal strain curve of the lithium silicate glass-ceramic doped with the quartz seed. a** Thermal strain curves of the composites with various quartz seed concentrations at 200–300°C. The inset describes the relationship between the coefficient of thermal expansion of the modified glass-ceramics and the quartz seed concentration. **b** Comparison of the thermal strain curves between the glass-ceramics with and without the addition of the quartz seed. The red box highlights a step-like change in the curve due to α-β cristobalite transition. **c** The phase composition of each original and modified glass-ceramics with the quartz seed concentrations of 3wt%, 5wt%, 7wt%, 9wt%. **d** The quartz seed content dependence of the total content of quartz

and quartz-cristobalite ratio. The dotted line represents the content of quartz seed, that's to say, the vertical height between the dotted line and yellow line denotes the extra quartz formed during the sintering process. **e** The average radius of the cylindrical sample with different quartz addition after the heat treatment. The black bar in the inset stands for 1 cm. **f** The Scanning electron microscopy images of the original and modified glass-ceramics with the addition of 7 wt% quartz seed after the 4 vol% hydrofluoric acid (HF) corrosion. **g** The comparison of the sealing time consumed in this work and conventional method. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

2.3. Improved glass-ceramics enable the sealing component with higher reliability

The lower level of the thermal stress in the sealing component usually represents the higher reliability of the sealing part. Therefore, in order to reduce the thermal stress in the sealing component, there have been several approaches to relieve the stress accumulation in the sealing process, such as annealing at a temperature marginally larger than glass transition temperature and avoiding an excessive cooling rate. However, thermal stress caused by a thermal expansion mismatch are inevitable when the component works at elevated temperature, especially at 200–300 °C in this case.

To compare the performance of the improved and original sealing glass in the sealing component, finite element method is carried out to simulate the sealing components at the actual situation where the component will encounter a high temperature environment, up to 500 °C. The sealing component usually consists of three parts, the pin (innermost), glass/glass-ceramics (middle), and housing (outermost). And the model consists of four repeated sealing parts. Retrospectively, the main aim of this work is to alleviate the concentration of stresses in the sealing component for the displacive phase transformation of cristobalite phase. And whether the maximum stress in the sealing component are effectively reduced is one of our most concerned issues. Fig. 4a, b are the cross-section of the model and the color in the model represents the distribution of the von Mises stress over the improved and original sealing component at 100 °C, 200 °C, 300 °C, 400 °C and 500 °C respectively. On the whole, the stress over the sealing glass mainly concentrated in the bottom and top of the glass and this indicates that these places might become the source of sealing failures if the stress exceed a critical value. Two kinds of the sealing glass have the similar distribution of the thermal stress under 200 °C. Additionally, it can be clearly found that in the cross-section of the component, the color in the image corresponding to improved glass is much lighter than that of the original glass at above 200 °C. In other words, the level of thermal stress in the improved glass is much lower than the original at above 200 °C, which suggests that this improved sealing glass is more suitable for the electronic component at elevated temperature. Fig. 4c illustrated the maximum stress in the sealing component, and this value often is used to evaluate whether stress exceed the failure criteria of a material. The maximum stress of the sealing component has been decreased from 163.5 MPa to 127.4 MPa at 250 °C, and make the maximum stress curve of the sealing glass doped with quart seed become smoother compared with the original one, especially at the transformation temperature where the α-β cristobalite phase transition occurs. As the temperature eventually grows up to 500 °C, the maximum stress in both sealing components will be closely equal to ~165 MPa. Meanwhile, in Fig. 4d, the average stress of the sealing glass experiences an evolution mode similar to the maximum stress curve before 300 °C. Once the temperature is larger than 300 °C, the average stress of the improved and original sealing glass simultaneously increases, whereas the slope of the improved one is smaller than that of the original one. Eventually, when the temperatures of the components reach up to the maximum, the average stress of the improved and original sealing components become 48.0 MPa and 64.7 MPa, respectively.



Download: Download high-res image (695KB)

Download: Download full-size image

Fig. 4. **The improved sealing glass-ceramics significantly reduces the mismatch stress in the sealing component.** The distribution of the von Mises stress over the **an** improved and **b** original sealing component. Higher stress is seen near the interface between the glass-ceramic and metal materials, especially in the top and bottom of the sealing glass. **c** The maximum

stress and **d** the average stress in the sealing component. The red line and blue line represent the improved and original sealing glass-ceramic, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

In hermetic sealing, the gas tightness of a device is generally ensured through a chemical bond at the interface. Specifically, this process usually involves the dissolution of the glass into the oxide film on the metal surface. To further understand why stress reduction leads to an improvement in gas tightness, it is necessary to find the coefficients of thermal expansion of the various components in the system from the supplementary materials. Among them, the housing has the highest coefficient of thermal expansion, the glass has the next highest, and the pin has the lowest. This means that in a high-temperature environment, tensile stress will be generated between the housing and the glass. The tensile stress at the interface can cause the housing and the seal to gradually separate, leading to a decrease in gas tightness. Therefore, when we reduce the stress level in the system through a thermal expansion behavior offset strategy, it will also directly improve the gas tightness of the device.

Unlike compression sealing, the hermeticity of the matched sealing component mainly depends on the chemical bonding between the glass and oxide layer on the metal materials. To obtain the high-quality interface, 304L stainless steels were first oxidized at an appropriate condition, then the oxide film will chemically react with the sealing glass and finally realize a good connection with the glass-ceramic, as present in Fig. 5a. Most importantly, the interface evolution at elevated temperature is one of utmost concern problem and thus, the connection components, including the original and improved one, were placed at 400–700°C for 5h. The results showed that the delamination occurred when the original sample was heated up at 700°C for 5h. As shown in Fig. 5b and c, there still exists a significant crack near the interface in the original one even when the temperature of the thermal aging is lower than 700°C. Interestingly, in Fig. 5d and e, the improved one maintains the complete though the sample was heated up at 700°C for 5h. Consequently, the improved glass exhibits an excellent ability to maintain hermeticity at elevated temperatures compared to the original sealing glass-ceramic.

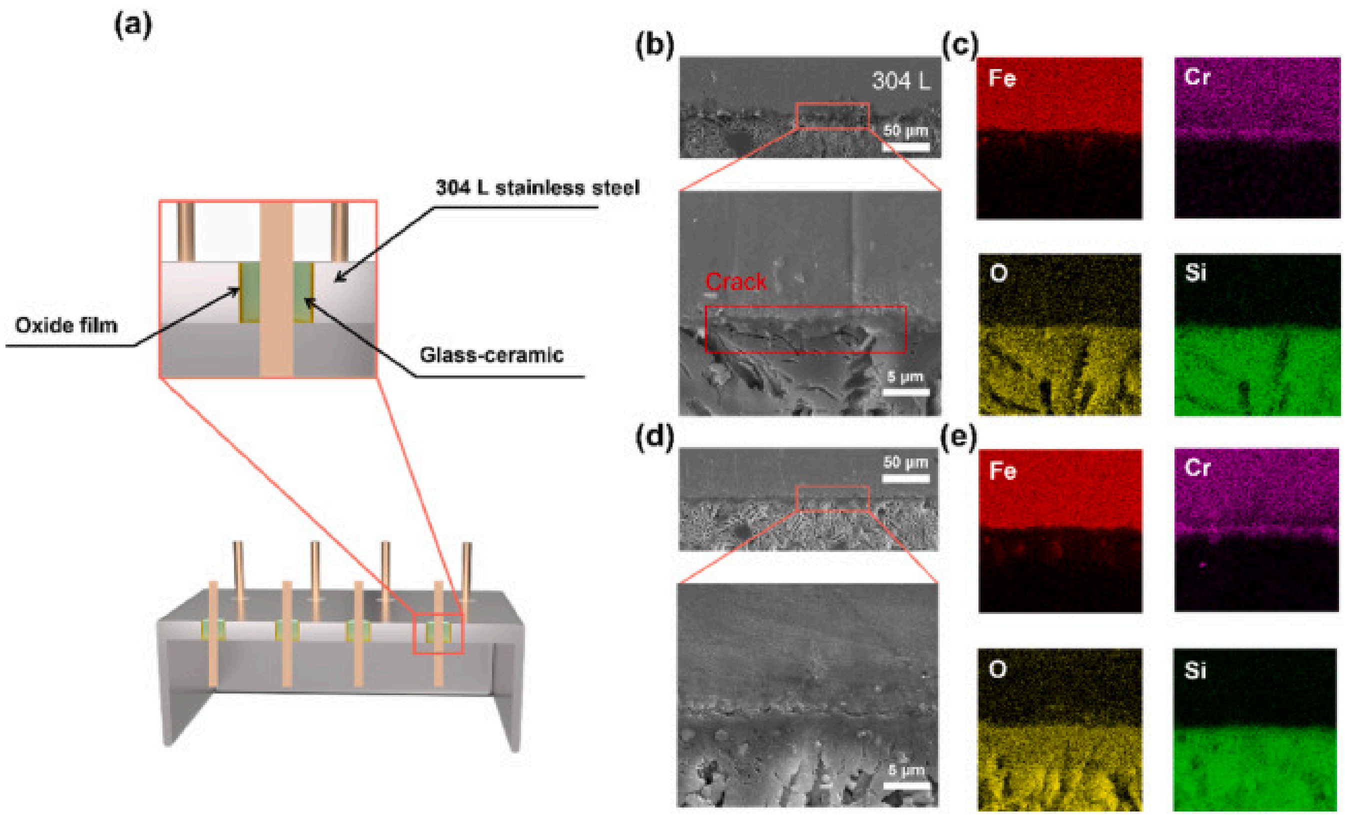


Fig. 5. **The enhanced stability at the interface of the sealing glass and metal. a** Diagram of the matched sealant and an enlarged fragmentary view of the matched sealant. **b** The microstructure and **c** element distribution of the sealant after the thermal aging (600°C, 5h). **d** The microstructure and **e** element distribution of the improved sealant after the thermal aging (700°C, 5h).

3. Conclusion

In the present work, we successfully demonstrate a thermal strain offset strategy for the fabrication of glass-ceramic to metal seals with high-reliability, and a novel high-expansion glass-ceramics (>17.0ppm/°C) with near-linear thermal strain curve are fabricated, through a facile and high-efficient thermal profile. As a Chinese saying goes, four ounces can move a thousand pounds. It's surprisingly found that a small amount of quartz seed can trigger the occurrence of more quartz phase at a higher temperature and slower cooling rate. Meanwhile, an improved thermal treatment schedule is designed to realize the matched hermetic seals in high-expansion alloys, such as 304L stainless steel owing to this exciting finding. Impressively, the sealing time consumed in the improved thermal treatment have been reduced, approximately 55 min, compared with the old sealing process, which extremely improve the production efficiency in the industry. And there is no need for additional equipment to achieve a rapid cooling rate in the improved thermal treatment process. The thermal strain behaviors at elevated temperature (0–500 °C) of the glass-ceramics with different quartz-cristobalite ratios is systematically investigated through a model based on the thermoelectricity, and thus the quartz-cristobalite ratio should be 1:1, with a goal to balance the high thermal expansion and linear degree of the thermal strain. In particular, the phase composition of the samples with the addition of 7 and 9 wt% quartz seed are consistent with the targeted phase compositions, whereas the glass-ceramics doped with 7 wt% quartz seed exhibits the better thermal expansion coefficient, and thus this glass-ceramic is considered as the optimized sealing glass. Significantly, the finite element simulation results of the component demonstrate that the improved sealing glass successfully ameliorate the mismatched stress caused by the cristobalite phase transformation. Clearly, the improved sealing glass will not only well enhance the reliability of the sealing component, but also largely reduce the sealing time and equipment cost in the sealing technology. Additionally, although in this work we have prepared a sealing glass with an ultra-high coefficient of thermal expansion through the thermal strain offset strategy, there is still insufficient systematic work on the stability of the improved glass-ceramic and the devices in extreme environments. And, exploring the use of ultra-high thermal expansion glass-ceramics in medium and low-temperature solid oxide fuel cells will be a worthwhile endeavor.

4. Methods

4.1. Materials

The purity degree of the raw materials and chemical formulates were recorded in the brackets. Silicon oxide (SiO_2, AR), zinc oxide (ZrO_2, AR), potassium carbonate (Na_2CO_3, AR), and boric acid (H_3BO_3, AR) was purchased from Sinopharm Chemical Reagent Co., Ltd. Alumina (MgO, AR), lithium phosphate (Li_3PO_4, 99%), lithium carbonate (Li_2CO_3, ≥99%) were bought from Shanghai Aladdin Biochemical Technology Co., Ltd.

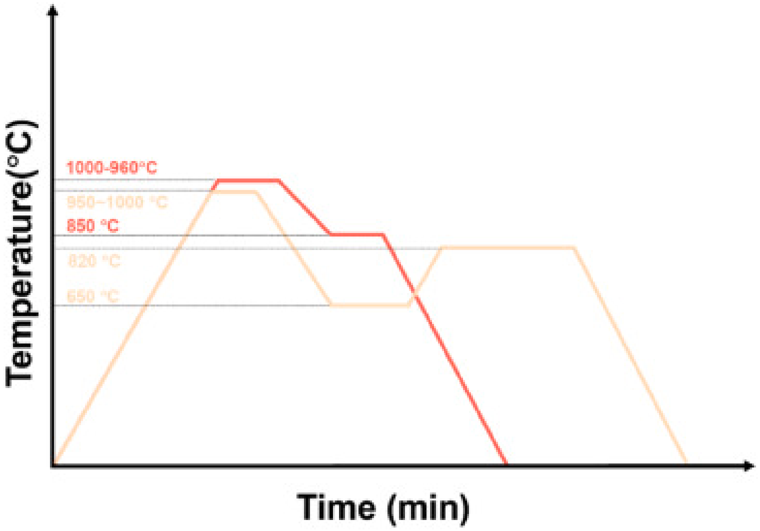
4.2. Material fabrication

The water-quenched method was employed to obtain the lithium silicate glass powders. The raw materials were weighted according to the chemical composition as shown in Table 1 and the weighted materials were homogenously effectively blended with zirconia balls by a pot mill machine for 12 h. Next, the mixed materials were placed into the platinum crucible and heated to 1600 °C for 3 h in a muffle furnace to ensure the uniformity and clarification of the melted glass liquid. The uniform and clarified glass liquid were poured into the deionized water to form the amorphous glass slag once the heat treatment had finished. Eventually, the glass slag transformed into the particles with a proper size, by ball milling for 30 min.

Table 1. The nominal chemical composition of the recrystallization lithium silicate glass-ceramics.

Raw materials	**Composition (wt.%)**
SiO_2	75.01
Li_2O	14.61
P_2O_5	4.15
MgO	2.80
Na_2O	1.95
B_2O_3	1.20
ZrO_2	0.28

The fabrication process of the samples used for characterization of the thermal strain curve consist of three steps in the whole process. The first step was the granulation of the original glass powder to increase the mobility of the particle. Subsequently, the granulation powders were pressed at 15 MPa and then the green body of the sample firstly heated to 475 °C, holding for 4 h to eliminate the organics in the green body and next to 870 °C for 30 min to get the relative densified sealing precursors. In the last steps, the sample would go through the sealing thermal profiles and precipitated some specific crystalline. In details, the precursors were placed inside the graphite fixtures with a shape of 3 mm × 4 mm × 50 mm slot for the heat treatment, and eventually the recrystallization lithium silicate glass-ceramics were obtained after the thermal profile, as shown in Fig. 6.



Download: Download high-res image (184KB)

Download: Download full-size image

Fig. 6. Sealing profile in this work (red line) and conventional method (yellow line). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

4.3. Materials characterization

Similarly, the detailed information, like a type, manufacturer, and country, were attached in the brackets. The thermal expansion strain of the samples was characterized by a dilatometer (DIL 402 Expedis Classic, NETZSCH, Germany). The sample used for the characterization of the thermal expansion strain were mold with a shape of 3 mm × 4 mm × 30 mm. The X-ray diffraction patterns were tested by a diffractometer (SmartLab, Rigaku, Japan), with a Cu Kα (λ = 1.5406 Å) radiation. High-temperature XRD of the samples were obtained at Bruker D8 ADVANCE (Germany) with a step of 0.02° and heating rate is 20 °C/min. Moreover, the bulk materials for the analysis of the phase composition were smashed into powders and then the powders of an average of 80 mesh fineness evenly mixed with 10 wt% alumina micro-powders. Finally, the quantitative analysis of the phase compositions in the glass-ceramics was performed in the Jade 6 software. The morphologies of the original and modified lithium silicate glass-ceramics were observed by a field emission scanning electron microscope (Mira 3, TESCAN, Czech). The nanoscale structures of the glass-ceramics were captured through a transmission electron microscope (Talo F200X, USA). The samples used for TEM test were prepared by focused ion beam. (FEI-FIB strata 400S, USA).

CRediT authorship contribution statement

Ming Huang: Writing – original draft, Investigation, Formal analysis, Methodology. **Fenglin Wang:** Investigation, Data curation, Conceptualization. **Haijun Mao:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Investigation. **Zhuofeng Liu:** Formal analysis, Data curation. **Wei Li:** Methodology, Investigation. **Weijun Zhang:** Writing – review & editing, Supervision, Resources, Project administration. **Xingyu Chen:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work is supported by the Natural Science Foundation of Hunan Province of China (Grant No. 2022JJ30661).

Appendix A. Supplementary data

The following is the Supplementary data to this article.

Download: Download Word document (1MB)

Multimedia component 1.

Recommended articles

Data availability

Data will be made available on request.

References

[1] K. Matusita, J.D. Mackenzie
Improvement of chemical durability of high expansion glasses by ion exchange
J. Mater. Sci., 13 (5) (1978), pp. 1026-1030, 10.1007/BF00544697 ↗
View in Scopus ↗ Google Scholar ↗

[2] I.W. Donald, P.M. Mallinson, B.L. Metcalfe, *et al.*
Recent developments in the preparation, characterization and applications of glass- and glass–ceramic-to-metal seals and coatings
J. Mater. Sci., 46 (7) (2011), pp. 1975-2000, 10.1007/s10853-010-5095-y ↗
View in Scopus ↗ Google Scholar ↗

[3] I. Ritucci, K. Agersted, P. Zielke, *et al.*
A Ba-free sealing glass with a high coefficient of thermal expansion and excellent interface stability optimized for SOFC/SOEC stack applications
Int. J. Appl. Ceram. Technol., 15 (4) (2018), pp. 1011-1022, 10.1111/ijac.12853 ↗
View in Scopus ↗ Google Scholar ↗

[4] K. Van der Wal, R.A.J. Van Ostayen, S.G.E. Lampaert
Ferrofluid rotary seal with replenishment system for sealing liquids
Tribol. Int., 150 (2020), Article 106372, 10.1016/j.triboint.2020.106372 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[5] X liang Dong, F. Zhou, W. Jiang, *et al.*
Failure analysis of the conical gaskets in ultra-high pressure pipeline sealing devices: experiment and numerical simulation
Eng. Fail. Anal., 133 (2022), Article 105946, 10.1016/j.engfailanal.2021.105946 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[6] K. Gong, Y. Cai, Z. Liu, *et al.*
Thermal cycle stability of glass-to-metal seals with glass preforms produced via powder-metallurgy and casting-machining methods
Mater. Res. Express, 10 (2) (2023), Article 025201, 10.1088/2053-1591/acb641 ↗
View in Scopus ↗ Google Scholar ↗

[7] I.V. Tolstobrov, E.S. Shirokova, A.I. Vepreva, *et al.*
Fused deposition modeling of glass sealants: a new approach to SOFC sealing
Ceram. Int., 50 (11, Part B) (2024), pp. 19561-19570, 10.1016/j.ceramint.2024.03.068 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[8] D. Lei, Z. Wang, J. Li, *et al.*
Experimental study of glass to metal seals for parabolic trough receivers
Renew. Energy, 48 (2012), pp. 85-91, 10.1016/j.renene.2012.04.033 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[9] C.H. Kuo, P. Cheng, C. Chou
Matched glass-to-Kovar seals in N_2 and Ar atmospheres
Int. J. Miner. Metall. Mater., 20 (9) (2013), pp. 874-882, 10.1007/s12613-013-0809-1 ↗
View in Scopus ↗ Google Scholar ↗

[10] Z hao Xiao, X yuan Sun, K. Liu, *et al.*
Crystallization behaviors, thermo-physical properties and seal application of Li_2O–ZnO–MgO–SiO_2 glass-ceramics
J. Alloys Compd., 657 (2016), pp. 231-236, 10.1016/j.jallcom.2015.10.106 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[11] C. Thieme, M. Schlesier, E. Oji Dike, *et al.*
Variable thermal expansion of glass-ceramics containing $Ba_{1-x}Sr_xZn_2Si_2O_7$
Sci. Rep., 7 (1) (2017), p. 3344, 10.1038/s41598-017-03132-x ↗
View in Scopus ↗ Google Scholar ↗

[12] A. Hunger, G. Carl, A. Gebhardt, *et al.*
Ultra-high thermal expansion glass–ceramics in the system $MgO/Al_2O_3/TiO_2/ZrO_2/SiO_2$ by volume crystallization of cristobalite
J. Non-Cryst. Solids, 354 (52) (2008), pp. 5402-5407, 10.1016/j.jnoncrysol.2008.09.001 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[13] S. Kolay, P. Bhargava
Phase and microstructural evolution in lithium silicate glass-ceramics with externally added nucleating agent
J. Am. Ceram. Soc., 102 (12) (2019), pp. 7312-7328, 10.1111/jace.16682 ↗
View in Scopus ↗ Google Scholar ↗

[14] Y. Yan, Y. Zhou
Preparation of cristobalite and its thermal characteristics
F. Dong (Ed.), Proceedings of the 11th International Congress for Applied Mineralogy (ICAM), Springer International Publishing, Cham (2015), pp. 441-445
Crossref ↗ Google Scholar ↗

[15] F. Molaei, M.S. Moghadam, S. Nouri
Investigation of thermal properties of crystalline alpha quartz by employing different interatomic potentials: a molecular dynamic study
Phys. Earth Planet. In., 316 (2021), Article 106724, 10.1016/j.pepi.2021.106724 ↗
View PDF View article View in Scopus ↗ Google Scholar ↗

[16] S. Dai, M.A. Rodriguez, J.J.M. Griego
Sealing glass-ceramics with near linear thermal strain, Part I: process development and phase identification
J. Am. Ceram. Soc., 99 (11) (2016), pp. 3719-3725, 10.1111/jace.14364 ↗
View in Scopus ↗ Google Scholar ↗

[17] W.F. Hammetter, R.E. Loehman
Crystallization kinetics of a complex lithium silicate glass-ceramic

J. Am. Ceram. Soc., 70 (8) (1987), pp. 577-582, 10.1111/j.1151-2916.1987.tb05709.x ↗

View in Scopus ↗ Google Scholar ↗

[18] C.K. Lin, L.H. Huang, L.K. Chiang, *et al.*
Thermal stress analysis of planar solid oxide fuel cell stacks: effects of sealing design
J. Power Sources, 192 (2) (2009), pp. 515-524, 10.1016/j.jpowsour.2009.03.010 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[19] C. Wang, J jun Yang, W. Huang, *et al.*
Numerical simulation and analysis of thermal stress distributions for a planar solid oxide fuel cell stack with external manifold structure
Int. J. Hydrogen Energy, 43 (45) (2018), pp. 20900-20910, 10.1016/j.ijhydene.2018.08.076 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[20] X. Wang, Z. Kou, H. Wu, *et al.*
Improvement of thermal cycle stability of YSZ-glass composite seals for intermediate temperature solid oxide fuel cell
Ceram. Int., 49 (22) (2023), pp. 36734-36742, 10.1016/j.ceramint.2023.08.359 ↗
Part B

View PDF View article View in Scopus ↗ Google Scholar ↗

[21] Y. Cao, G.C. Xu, E. Zanchi, *et al.*
Electrochemically enhanced wetting of 3YSZ and Mn1.5Co1.5O4 by Ag−CuO and their application in flash joining
Ceram. Int., 49 (22) (2023), pp. 35340-35348, 10.1016/j.ceramint.2023.08.207 ↗
Part A

View PDF View article View in Scopus ↗ Google Scholar ↗

[22] H. Tang, Y.C. Zhang, J.L. Zhang, *et al.*
Effect of holding time in brazing cooling process on creep life of the solid oxide fuel cell with bonded compliant seal

Int. J. Hydrogen Energy, 48 (58) (2023), pp. 22220-22230, 10.1016/j.ijhydene.2023.03.051 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

[23] T.J. Headley, R.E. Loehman
Crystallization of a glass-ceramic by epitaxial growth
J. Am. Ceram. Soc., 67 (9) (1984), pp. 620-625, 10.1111/j.1151-2916.1984.tb19606.x ↗

View in Scopus ↗ Google Scholar ↗

[24] J.W.P. Schmelzer, O.V. Potapov, V.M. Fokin, *et al.*
The effect of elastic stress and relaxation on crystal nucleation in lithium disilicate glass
J. Non-Cryst. Solids, 333 (2) (2004), pp. 150-160, 10.1016/j.jnoncrysol.2003.10.002 ↗

View PDF View article View in Scopus ↗ Google Scholar ↗

Cited by (9)

Thermo-mechanical coupling in electrical penetration assembly: residual strain analysis with embedded FBG strain monitoring

2026, Optics and Laser Technology

Show abstract

Synergistic mechanism between interfacial bonding and geometry-driven compression in glass-to-metal seals

2026, Journal of Non Crystalline Solids

Show abstract

Effect of precompression on the thermal cycling stability of glass-to-metal seals

2025, Materialia

Show abstract

Enhanced sealing properties of LiF-SrO-Bi2O3-SiO2 glasses via MgO/SrO replacement and controlled crystallization
2025, Ceramics International

Show abstract

In situ monitoring of cristobalite within a glass matrix via high-temperature X-ray Diffraction (XRD) ↗
2025, Powder Diffraction

Achieving near-linear thermal expansion and in situ interfacial regulation in Li2O–ZnO–SiO2 sealing glass–ceramics ↗
2026, Journal of the American Ceramic Society

View all citing articles on Scopus ↗

View Abstract